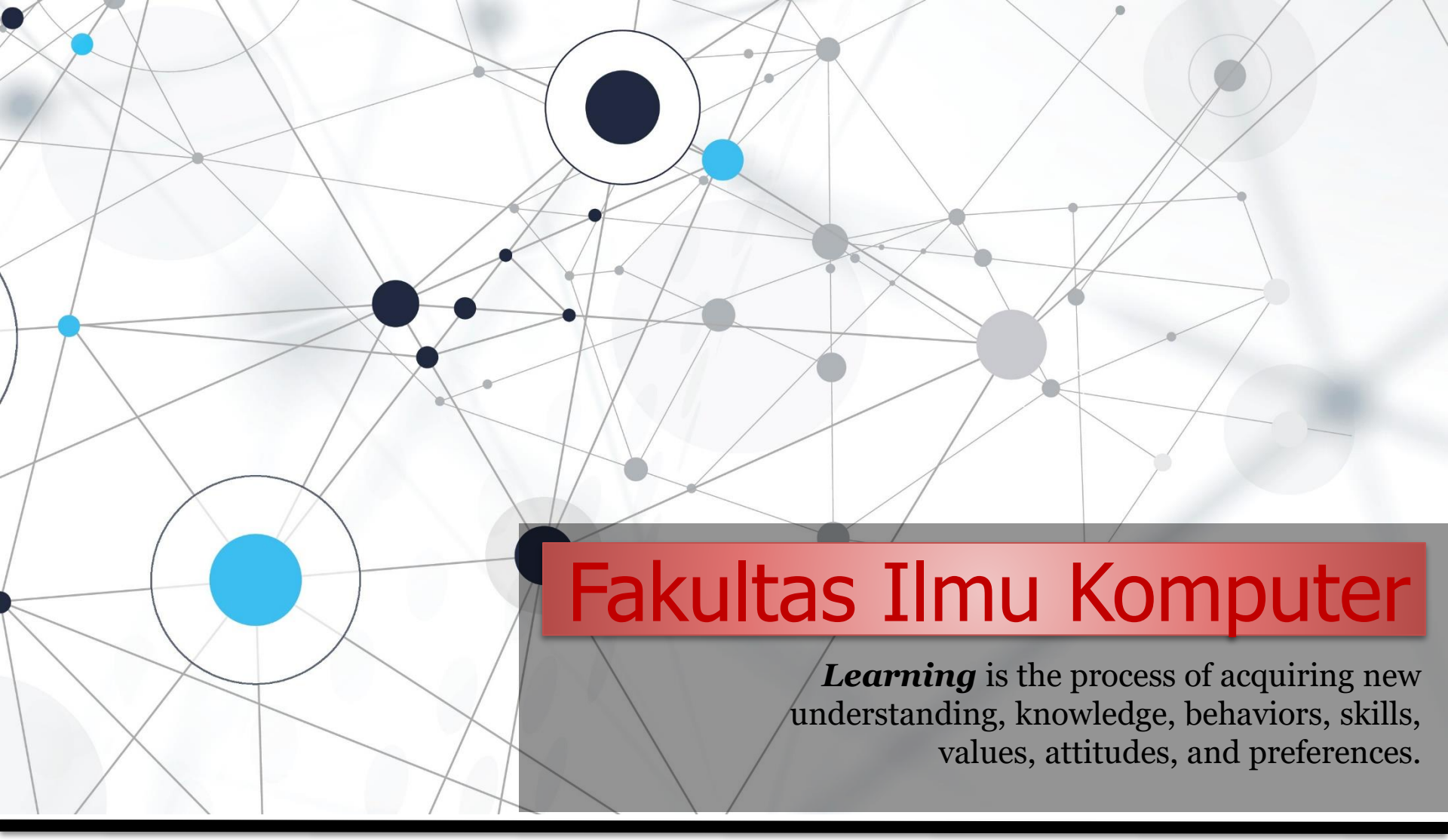
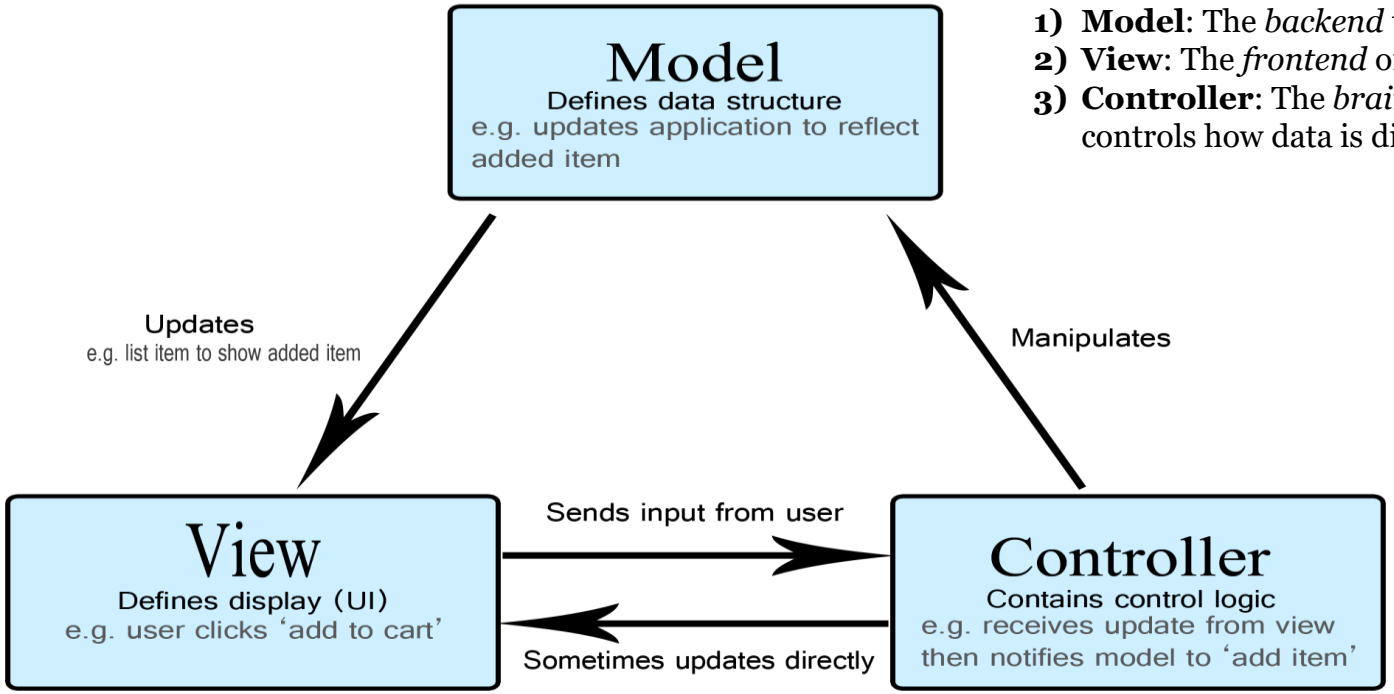


Default Controller & Method

Defining a Default Controller and Method

What is a Controller?



1) **Model**: The *backend* that contains all the data logic.
2) **View**: The *frontend* or graphical user interface (GUI).
3) **Controller**: The *brains* of the application that controls how data is displayed.

The Model
The model defines what data the app should contain. If the state of this data changes, then the model will usually notify the view (*so the display can change as needed*) and sometimes the controller (*if different logic is needed to control the updated view*).

The View
The view defines how the app's data should be displayed.

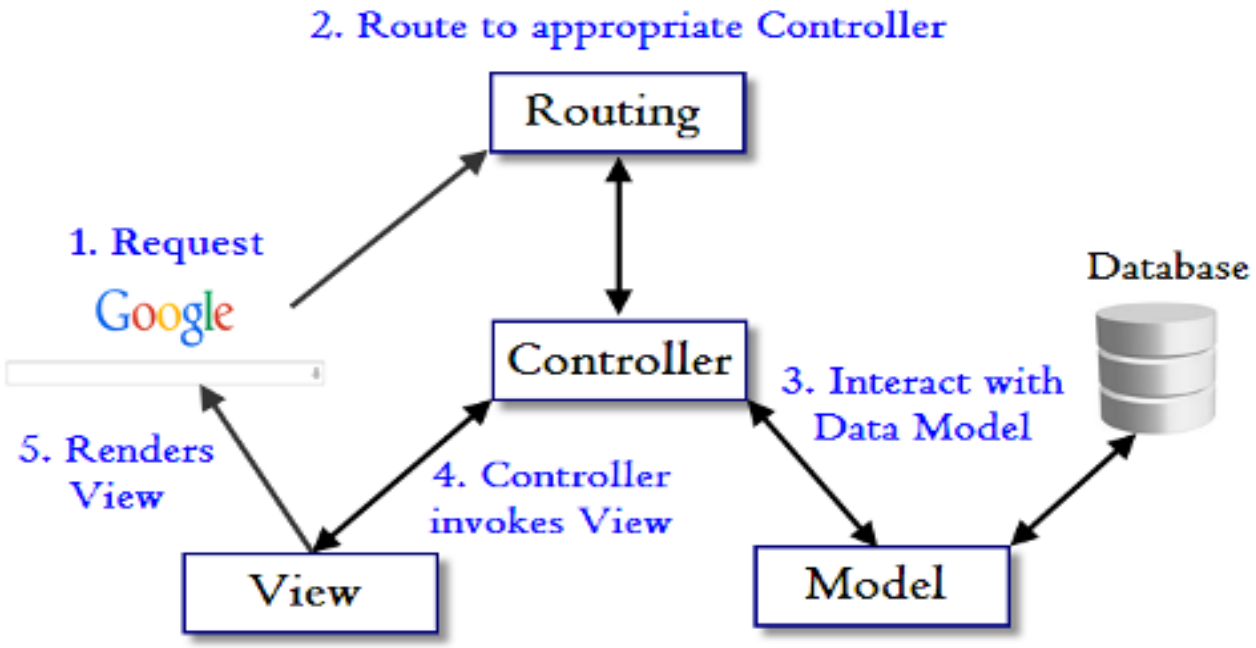
The Controller
The controller contains logic that updates the model and/or view in response to input from the users of the app.

What is a Controller?

Controllers are the *heart/brain* of our application, as they determine how HTTP requests should be handled.

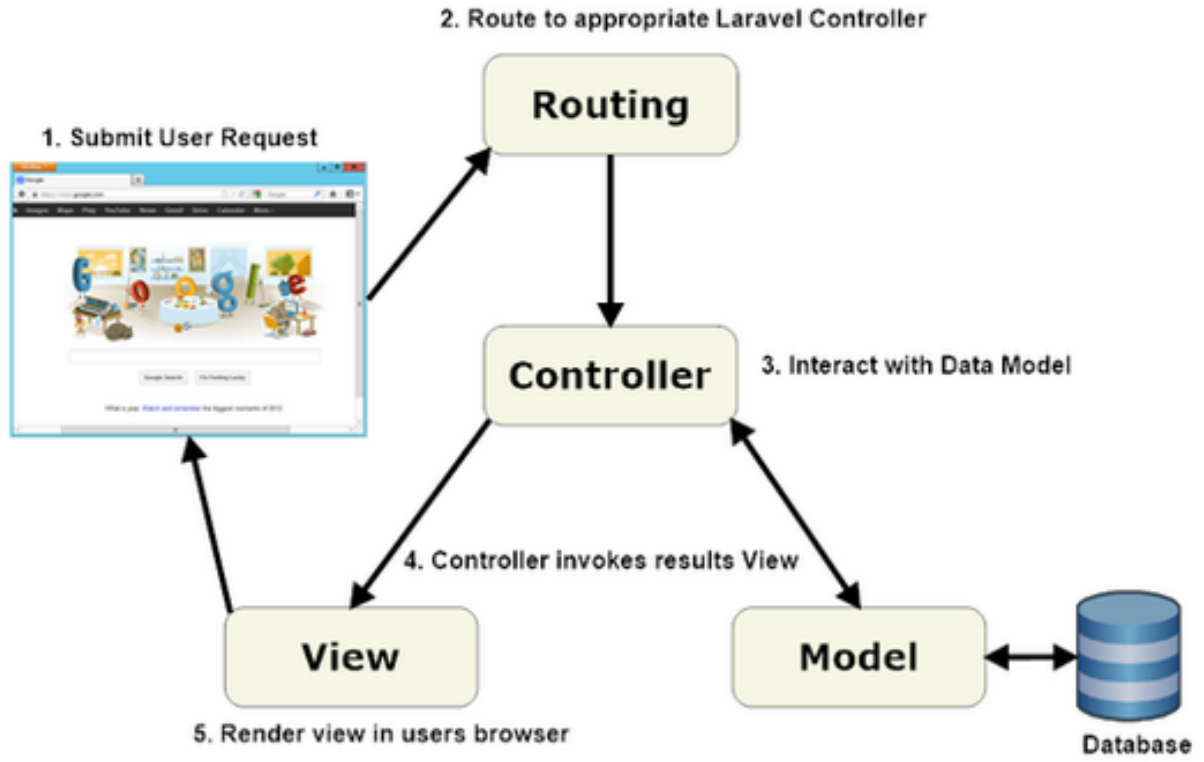
A ***Controller*** is simply a class file that is named in a way that can be associated with a URI.

What is a Controller?



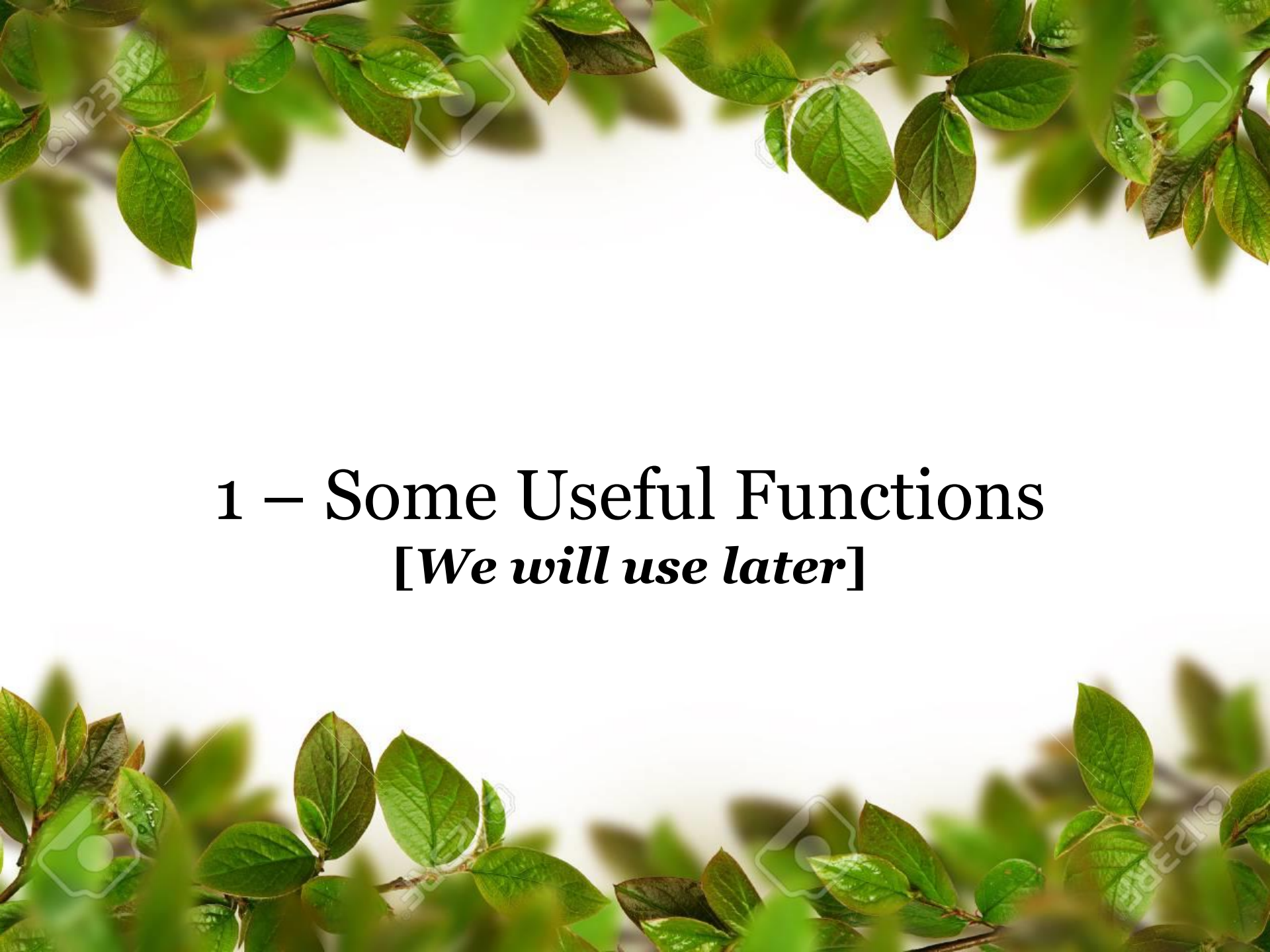
Normally a *user* interacts with the ***view*** and performs some action on the screen. A *request* is generated from the *view* which is handled by the ***controller***. ***Controller*** is actually responsible for handling *http request*.

Laravel Controller



MVC URL Routing system allows to create powerful and flexible URLs for the applications.
It provides two functionalities:

1) **Inspect incoming URLs** - This functionality manifests when the application receives a client request.
2) **Generate outgoing URLs** - This functionality manifests when we render URLs using Url Action in the View.



1 – Some Useful Functions

[*We will use later*]

1 - PHP file_exists() Function

The file_exists() function checks whether a file or directory exists. **Note**: The result of this function is cached. Use clearstatcache() to clear the cache.

```
<?php

    // check whether file exists or not
    $file_pointer = '/user01/work/gfg.txt';

    if (file_exists($file_pointer)) {
        echo "The file $file_pointer exists";

    }else{
        echo "The file $file_pointer does not exists";
    }

?>
```

2 - PHP unset() Function

The unset() function destroys a given variable. Here we use unset() function to delete a given element index in array.

```
$a = array(1,2,3,4,5);
unset($a[2]);
print_r($a);

Array
(
    [0] => 1
    [1] => 2
    [3] => 4
    [4] => 5
)
```

Index 2 already removed.

3 - PHP method_exists() Function

It checks if the class method exists in the given object. It returns TRUE if the method given by method_name has been defined for the given object, FALSE otherwise.

```php
<!DOCTYPE html>
<html>
<body>
  <?php
    class Fruit {
      // Properties
      public $name;
      public $color;

      // Method set name
      function set_name($name) {
        $this->name = $name;
      }

      // Method get name
      function get_name() {
        return $this->name;
      }
    }

    // Create object
    $obj = new Fruit();

    if(method_exists($obj, 'get_name')){
      echo 'Method exist.';
    }else{
      echo 'Method not exist.';
    }
  ?>
</body>
</html>
```

```
Method exist.
```

Sr.No	Parameter & Description
1	**object(Required)** The tested object
2	**mthod_name(Required)** The method name.

4 - PHP array_values() Function

The array_values() function returns an array containing all the values of an array.
Tip: The returned array will have numeric keys, starting at 0 and increase by 1.

```php
// PHP function to illustrate the use of array_values()
function Return_Values($array)
{
    return (array_values($array));
}

$array = array("ram"=>25, "krishna"=>10, "aakash"=>20, "gaurav");
print_r(Return_Values($array));
?>
```

Output:

```
Array
(
    [0] => 25
    [1] => 10
    [2] => 20
    [3] => gaurav
)
```

5 - call_user_func_array() Function

The call_user_func_array() function call a user function given with an array of parameters. The call_user_func_array() function can call a custom function "*function*" with the parameters from "param_arr array".

Syntax

```
mixed call_user_func_array( callback function [, array param_arr])
```

```php
<?php
   function Box($width,$height, $depth) {
      $b = $width*$height*$depth;
      echo $b;
   }
   call_user_func_array("Box", array("width" => 10, "height" => 20, "depth" => 30));
?>
```

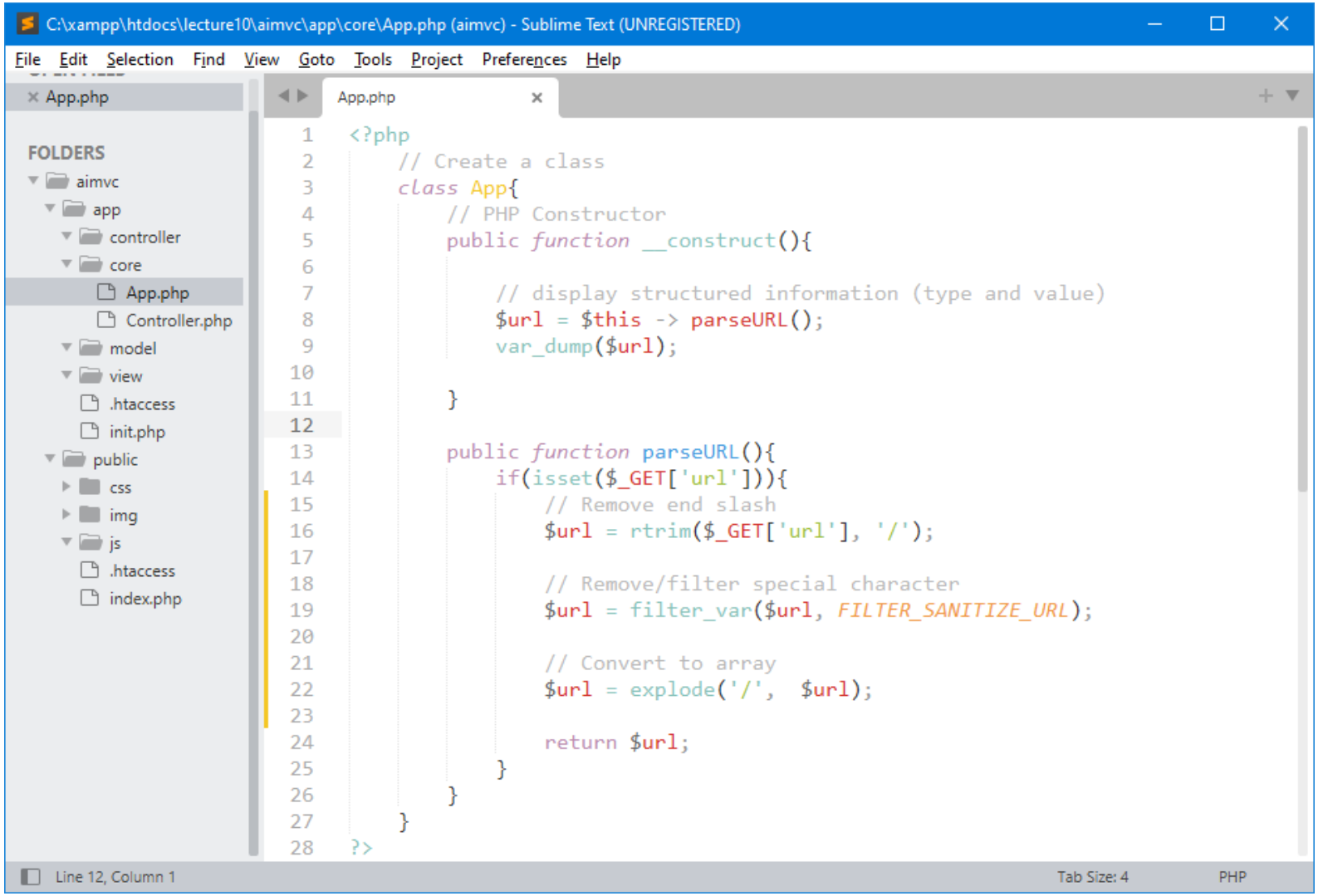
Output

```
6000
```

2 – Check Routing Rules from Previous Lecture

Handling Request URL

Handle more *routing rules* when *users* try to request through URL.



URL Routing

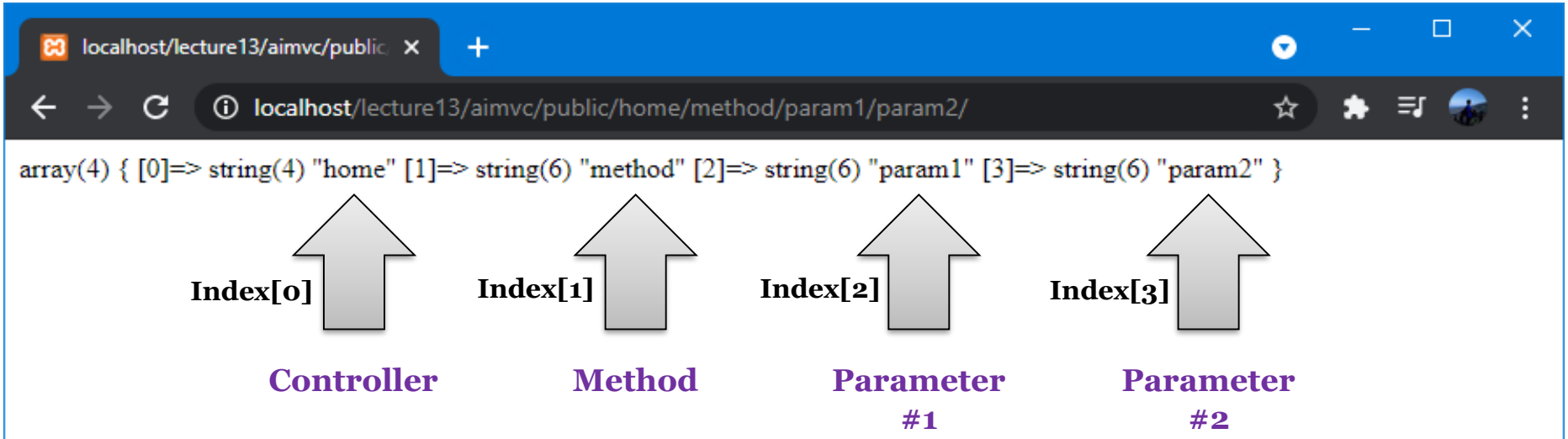
Typically there is a one-to-one relationship (*hubungan satu-ke-satu*) between a URL string and its corresponding controller class/method. The segments in a URL (*bagian dalam URL*) normally follow this pattern:

*http:/www.example.com/**class**/**method**/**parameter1**/**parameter2**/*

What is a Controller?
A Controller is simply a class file that is named in a way that can be associated with a URI.

Class merupakan *controller class* yang sudah dibuat dan biasanya segmen kedua (***method***) dari URL disiapkan untuk nama *metode*.

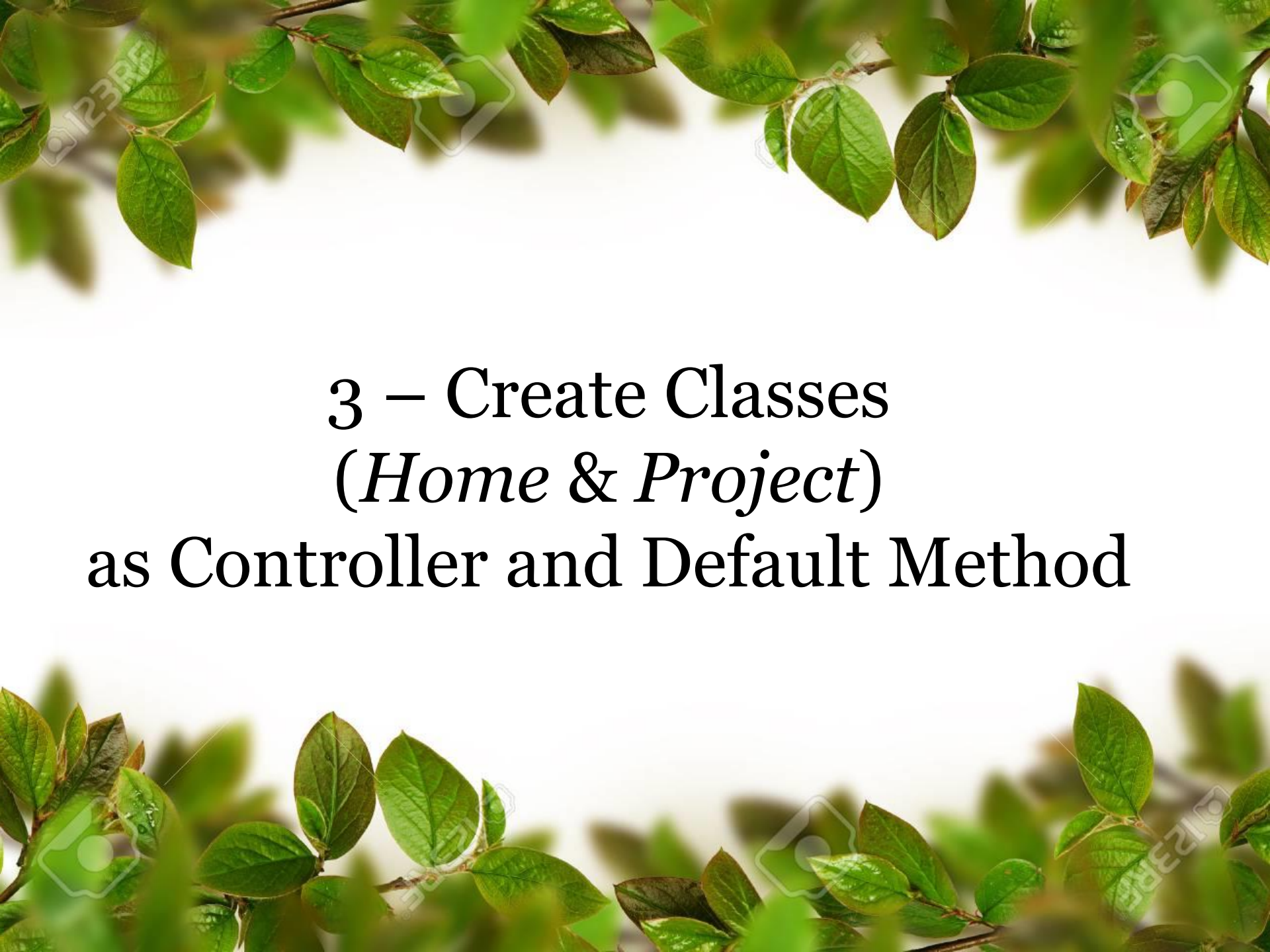
URI Routing from Previous Demo



Our app starts from this directory location:
http://localhost/lecture13/aimvc/public/

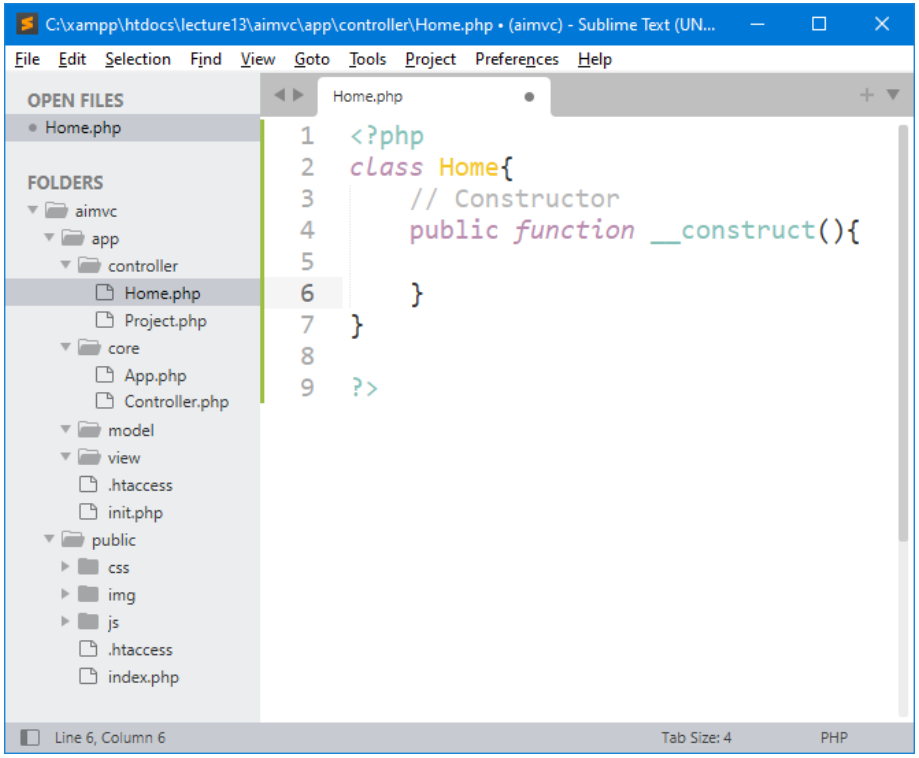
This means Controller default (home) and method default (index).

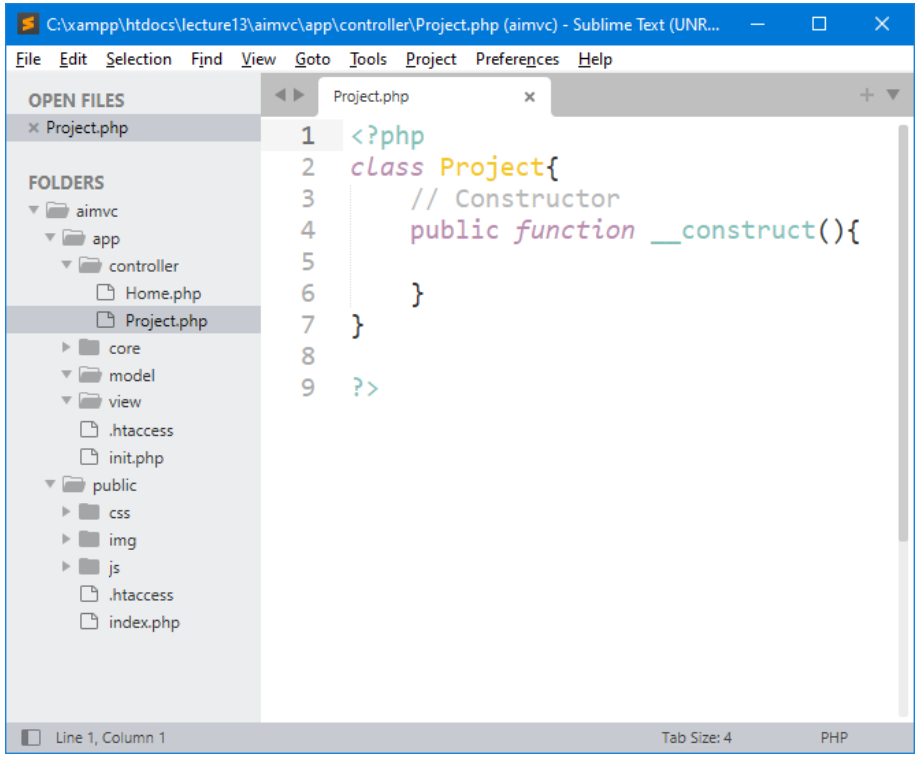
***URL Format*:** http://localhost/lecture13/aimvc/public/class/method/param1/param2/



3 – Create Classes (*Home* & *Project*) as Controller and Default Method

Controller Classes

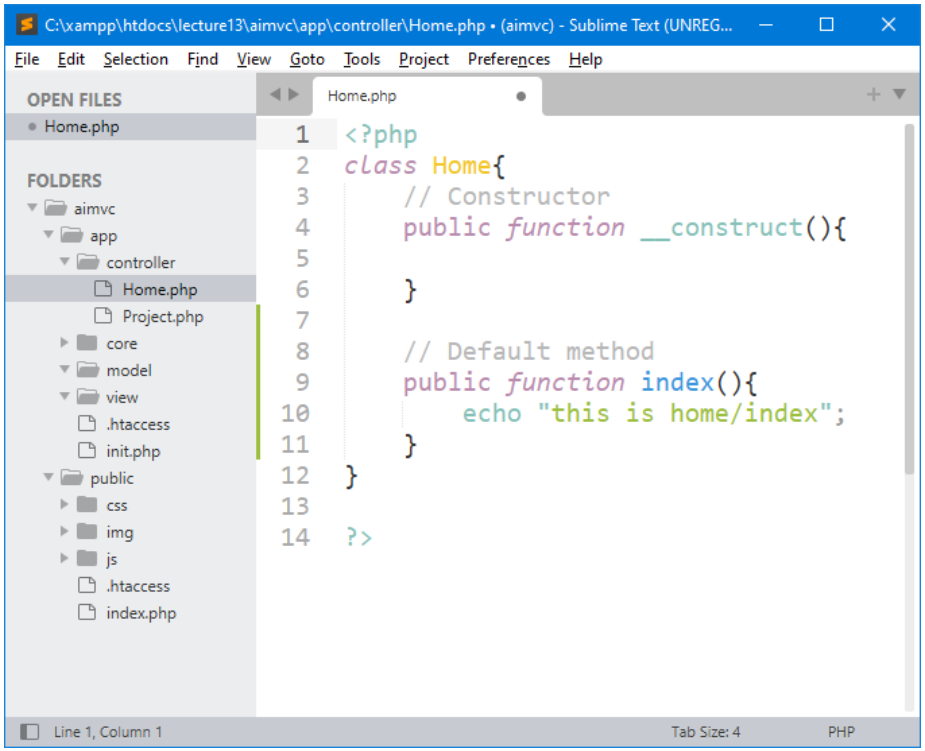


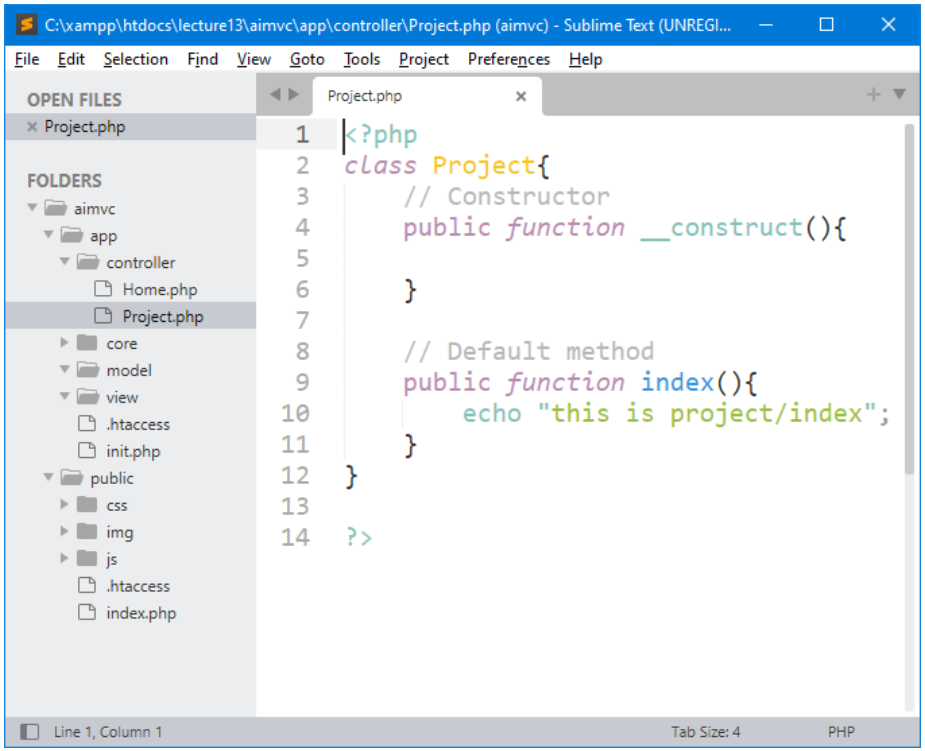


Important Note:
Class names must start with an uppercase letter.

Create *two classes* in *controller* directory to represent/handle ***home*** and ***project*** pages in our framework.

index() as a Default Method





When we do not write/request anything in the URL ("*/aimvc/public/*"), then this method (*index*) will be called.

___construct() vs index() methods

Why don't you just use the constructor instead of the index method?

__construct() is called first, then according to URL is called index() or other functions.

Public function __construct() should contain:

1) allocating resources used in entire class ex. $this->load
2) check user authentication (if entire class requires it)

Public function index() should contain:

1) allocating resources used only in this function
2) calling views or displaying anything

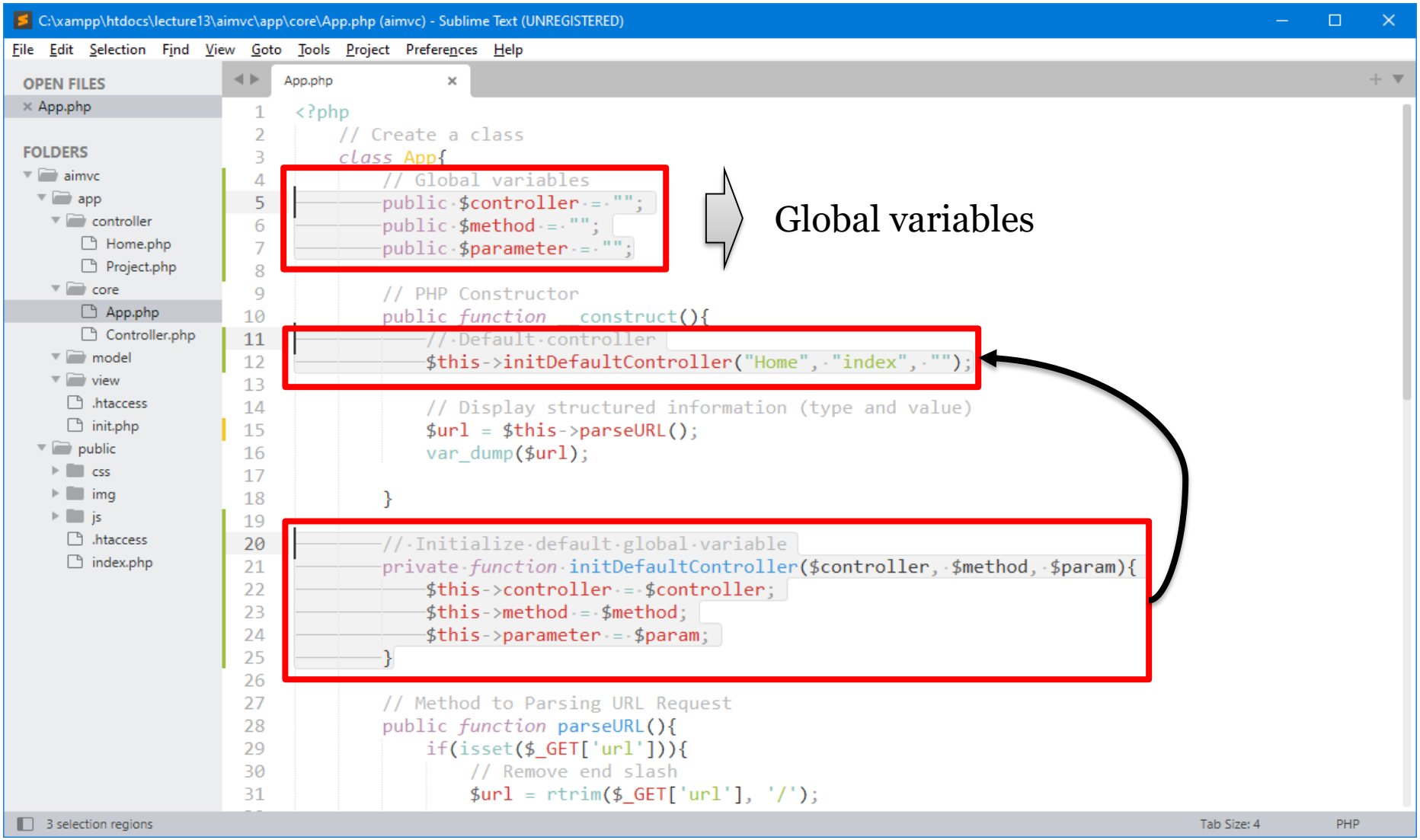
Link:

https://stackoverflow.com/questions/16691036/codeigniter-2-difference-between-index-and-construct-and-what-to-put-in-cons

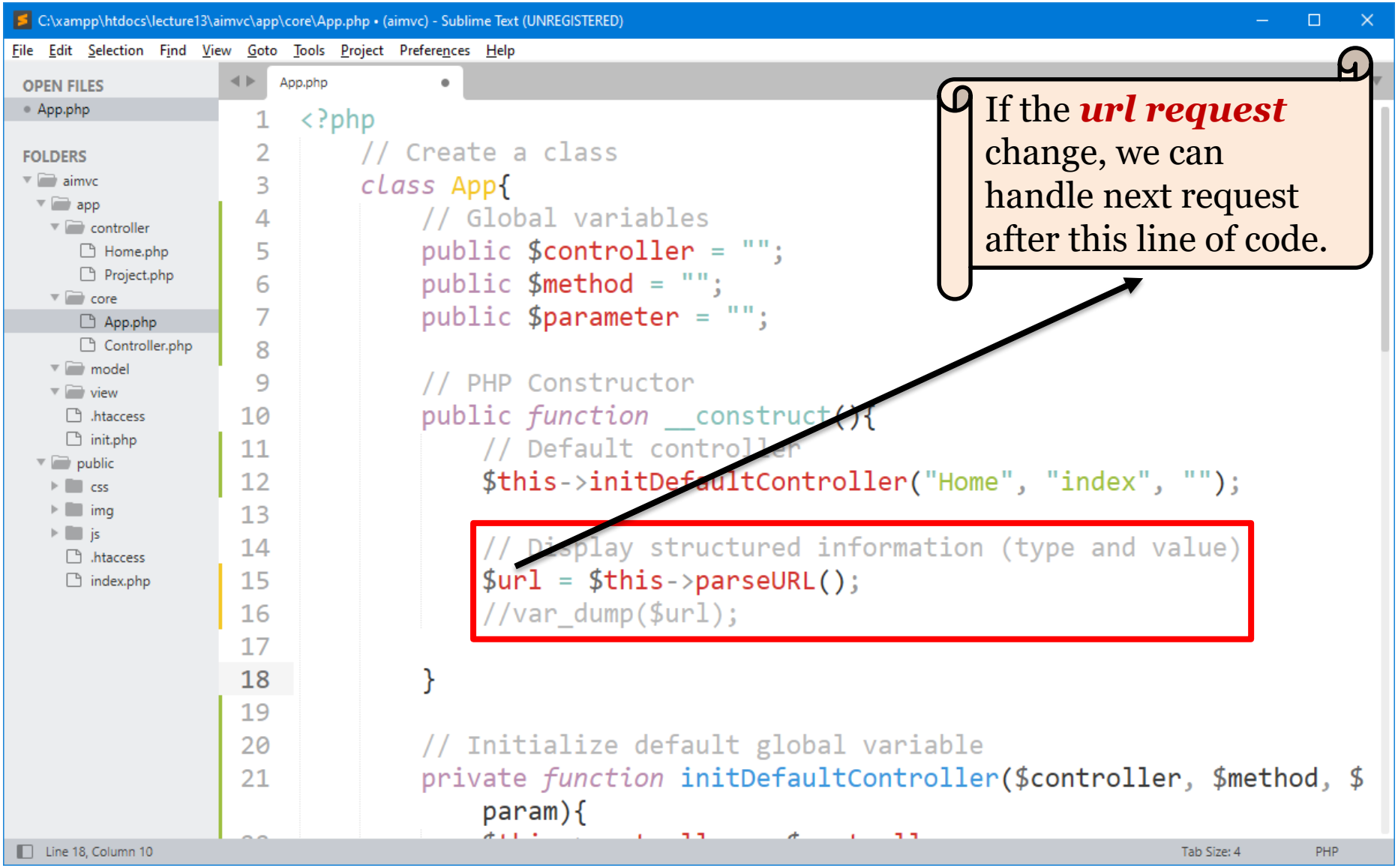
4 – Handle URL Request with Controller Classes (*home* & *project*)

Create Global Variables

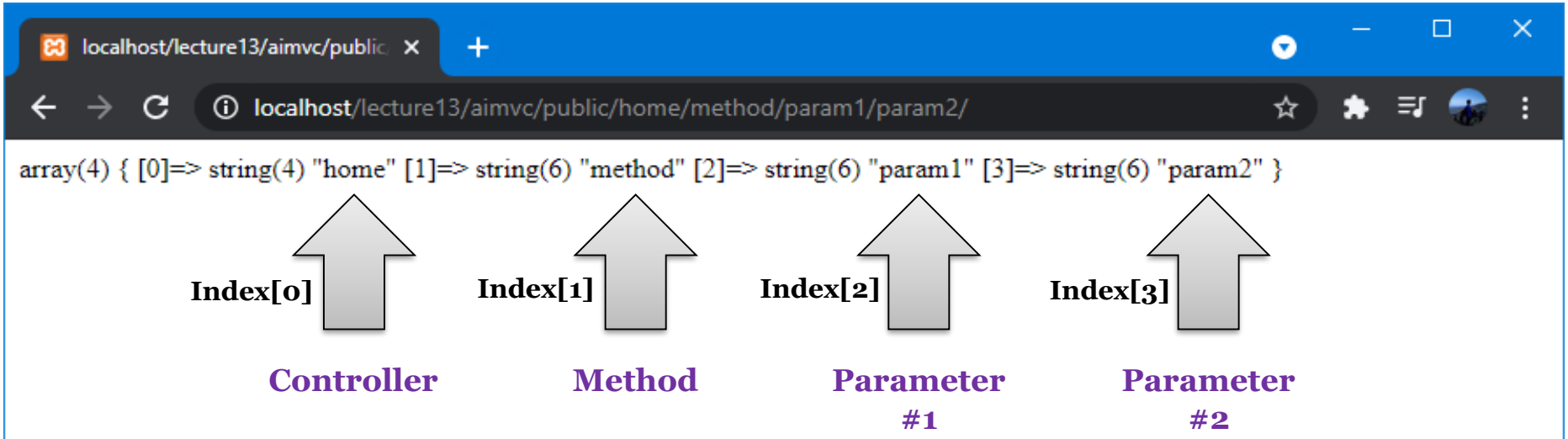
In PHP class, global variable as property in class to store a ***default controller (home)***.



Remember this $url variable !!!



After Add Some URI Routing Rules

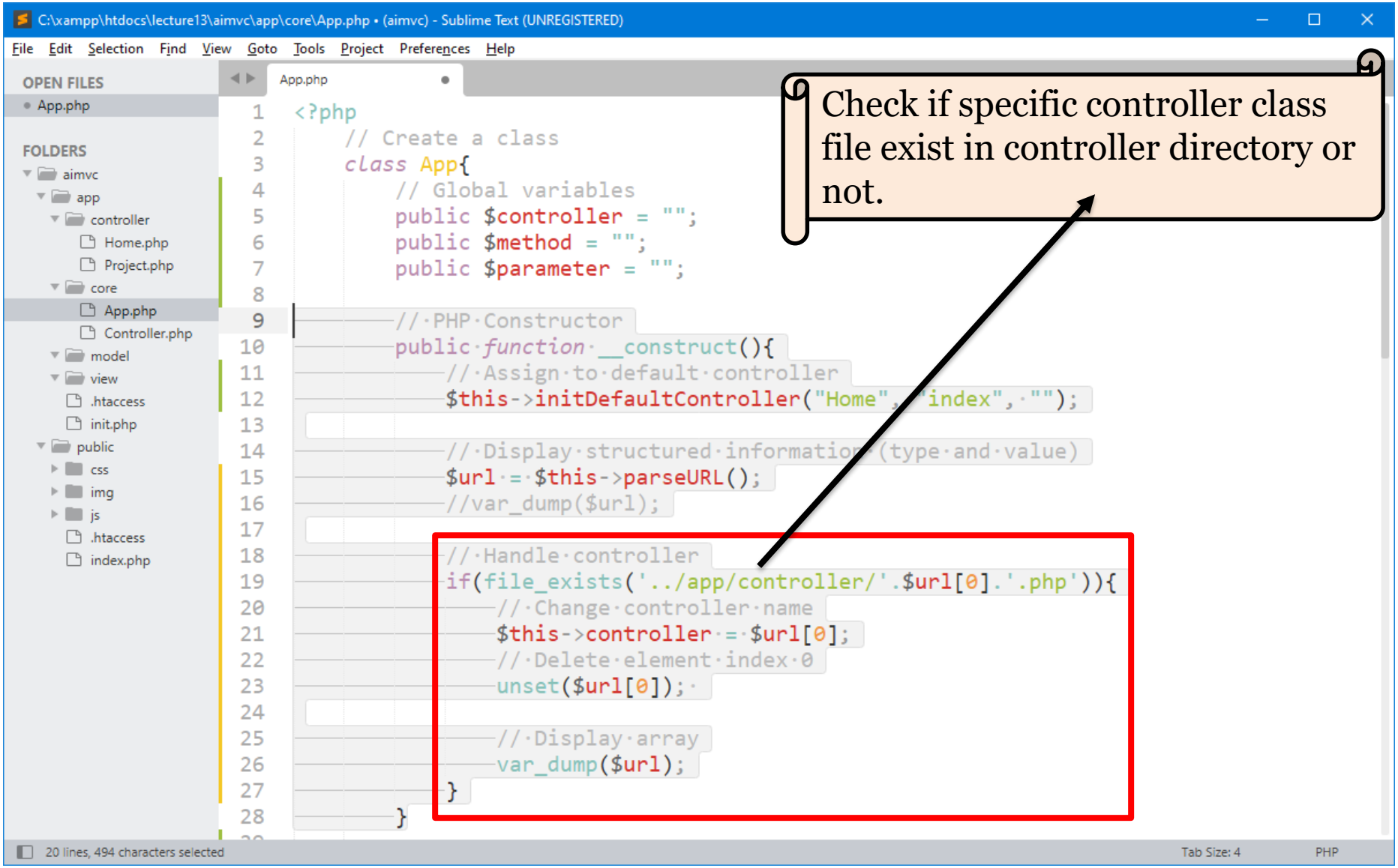


Our app starts from this directory location:
http://localhost/lecture13/aimvc/public/

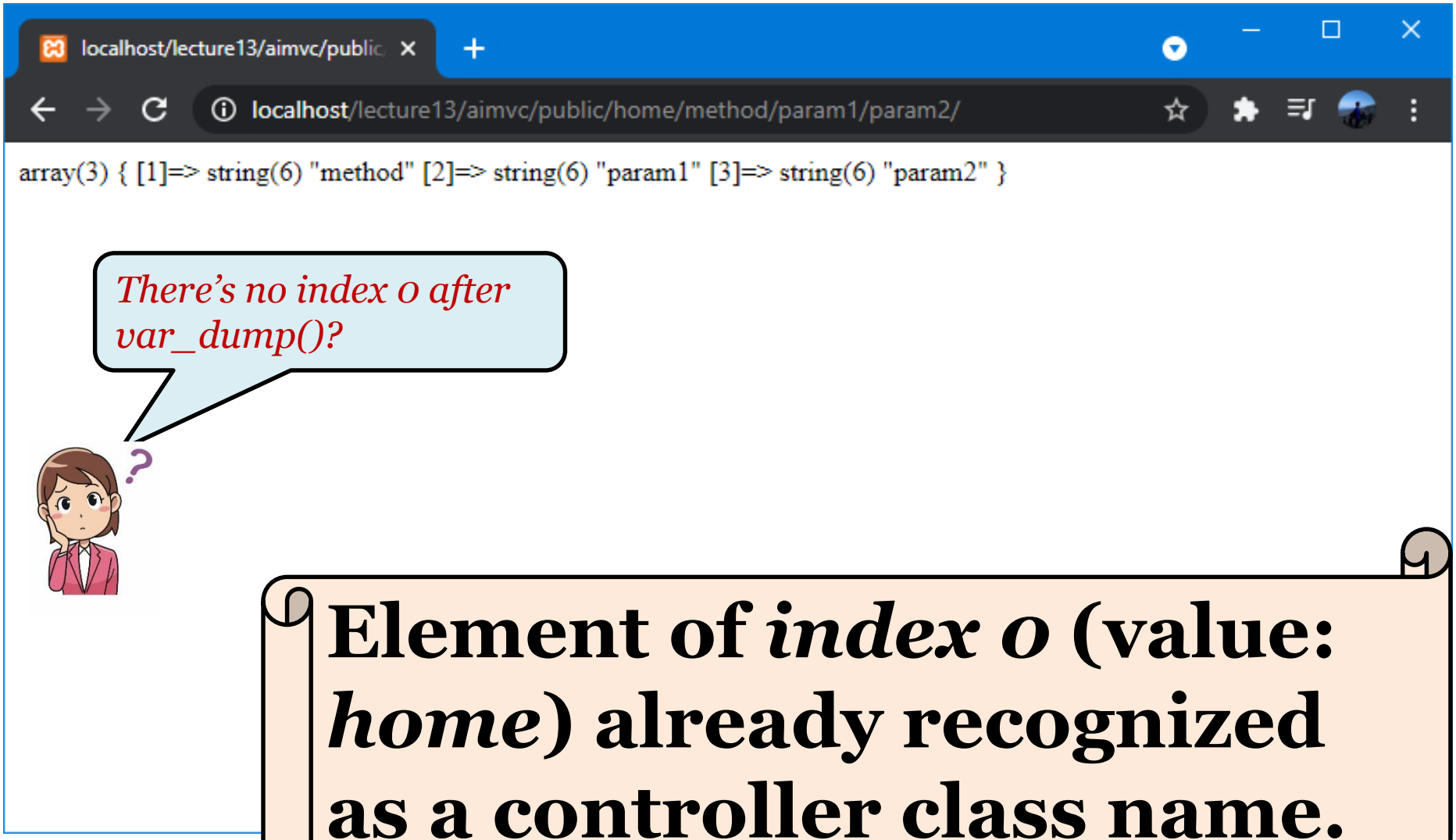
This means Controller default (*home*) and method default (*index*).

***URL Format*:** http://localhost/lecture13/aimvc/public/home/method/param1/param2/

Get element *index 0* as Controller Name

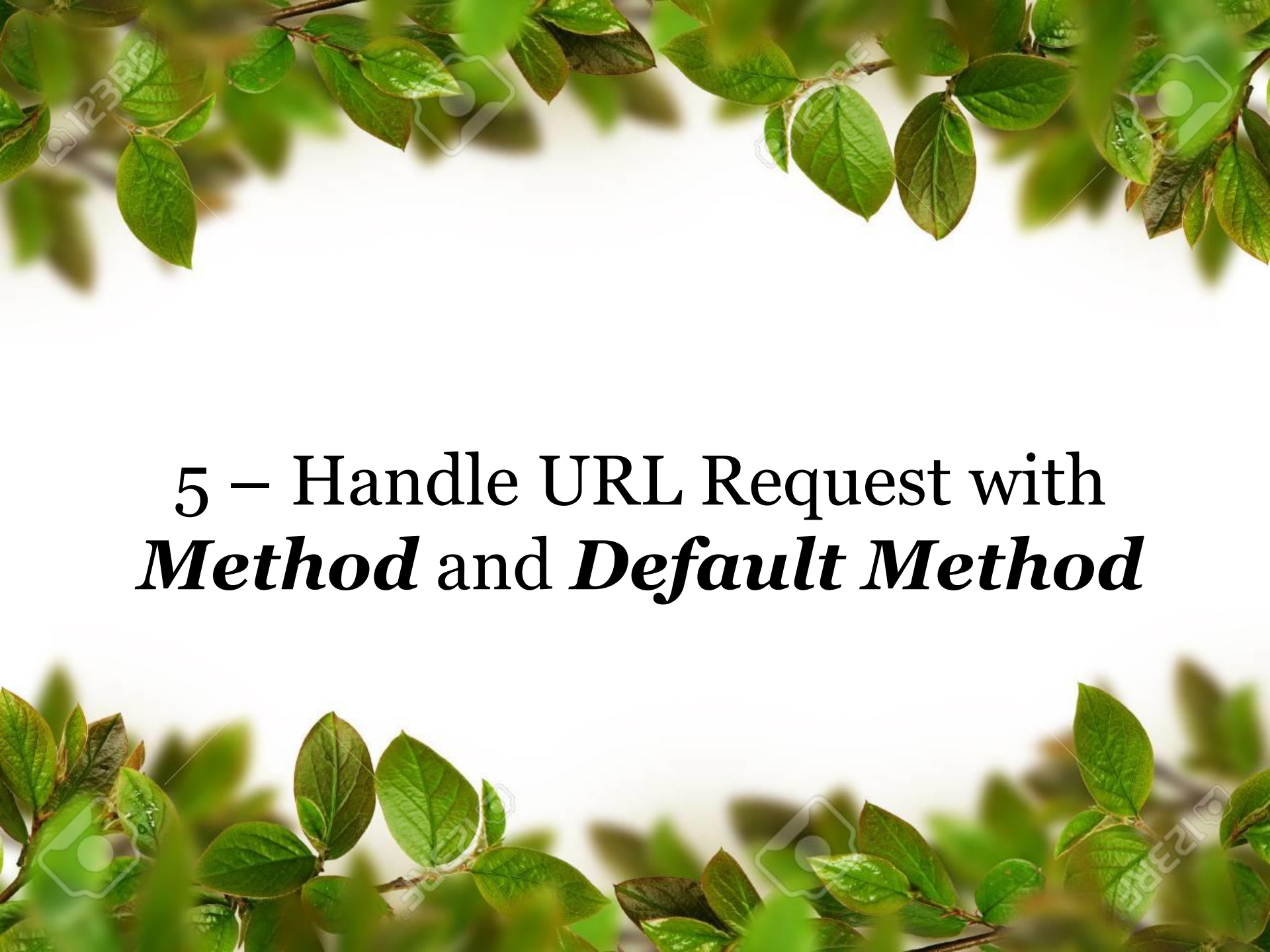


Recognize Controller Name



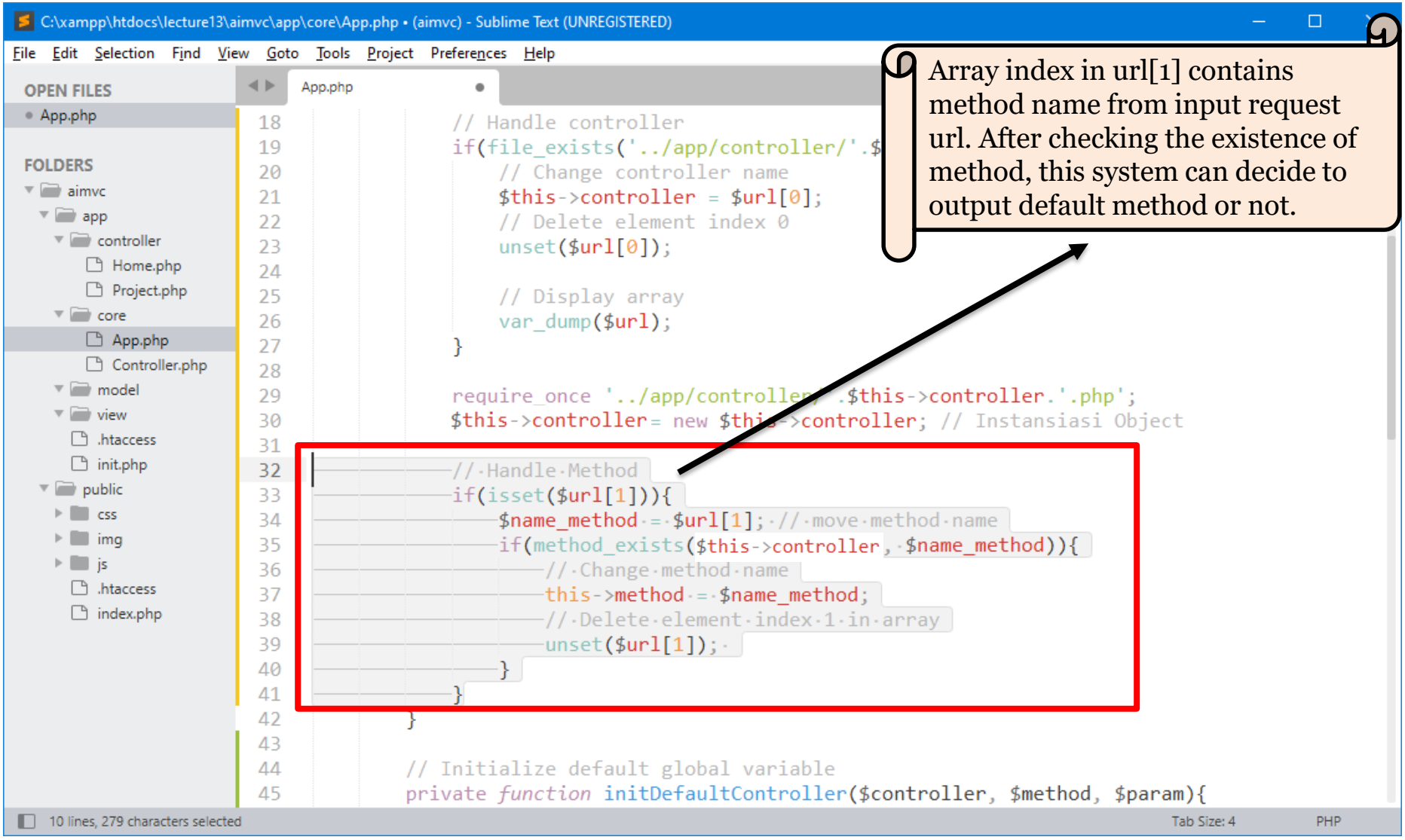
Element of *index 0* (value: *home*) already recognized as a controller class name.

Full Handle Controller

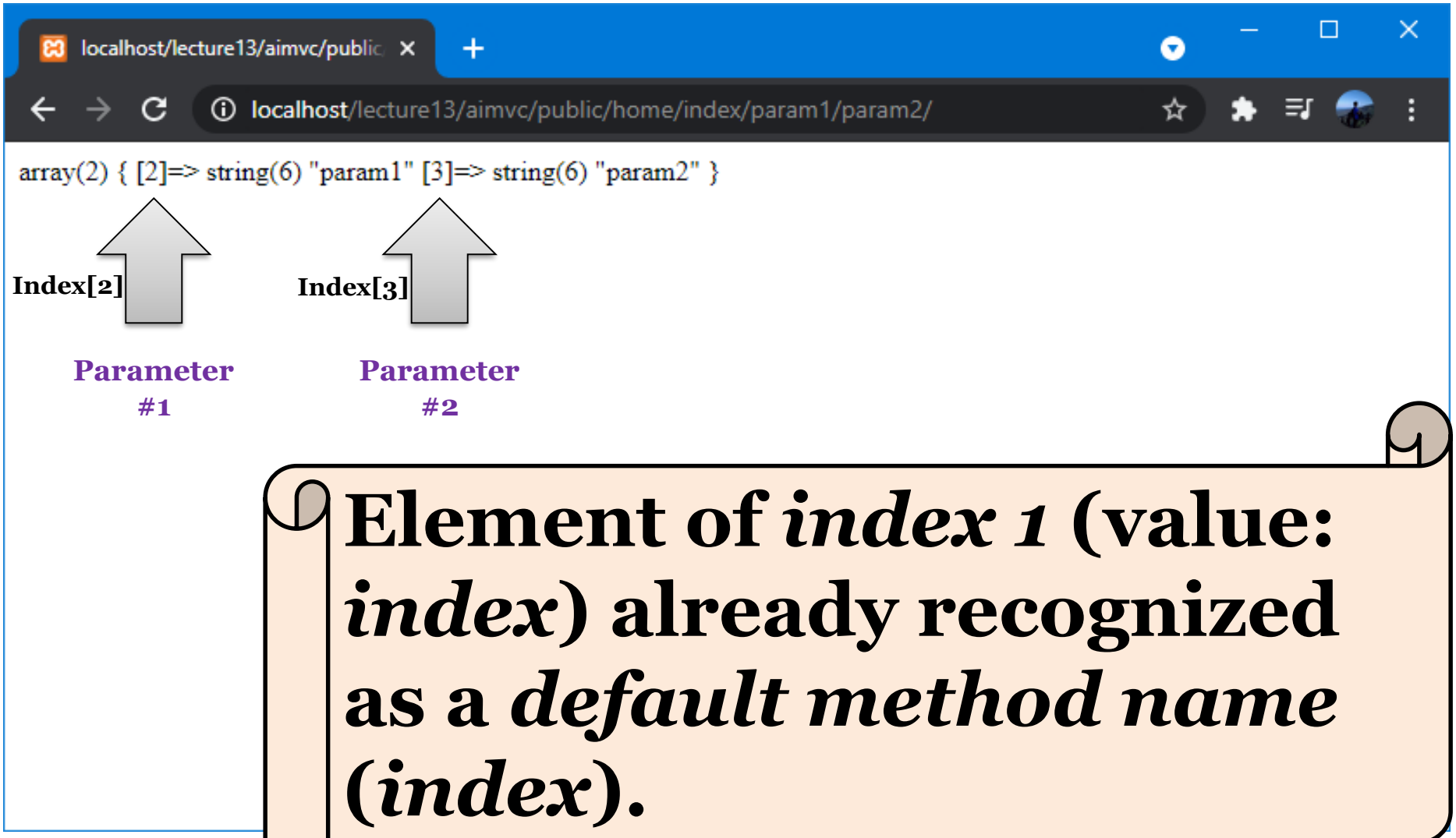


5 – Handle URL Request with ***Method*** and ***Default Method***

Recognize Method Name (default or not)



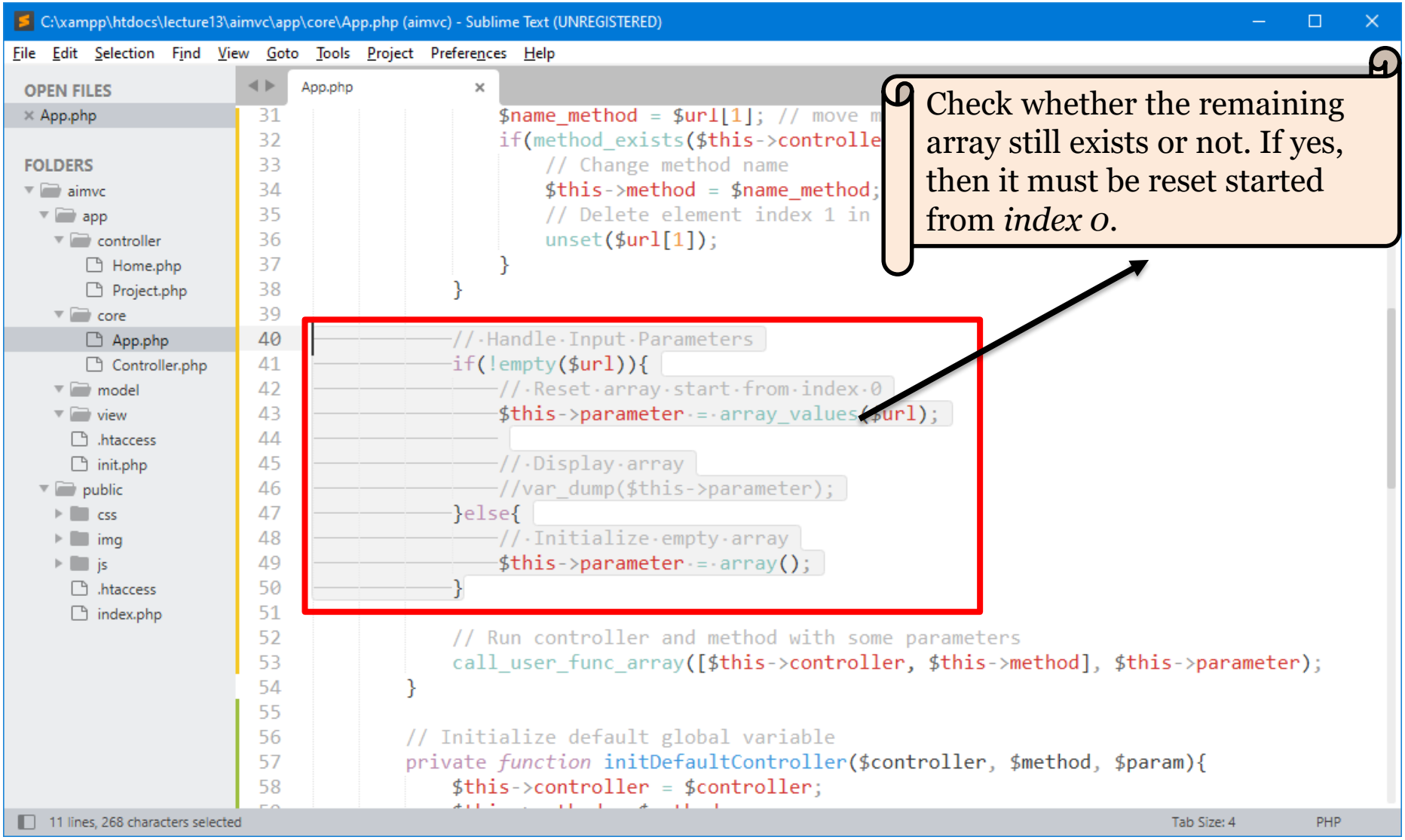
Test Handle Method



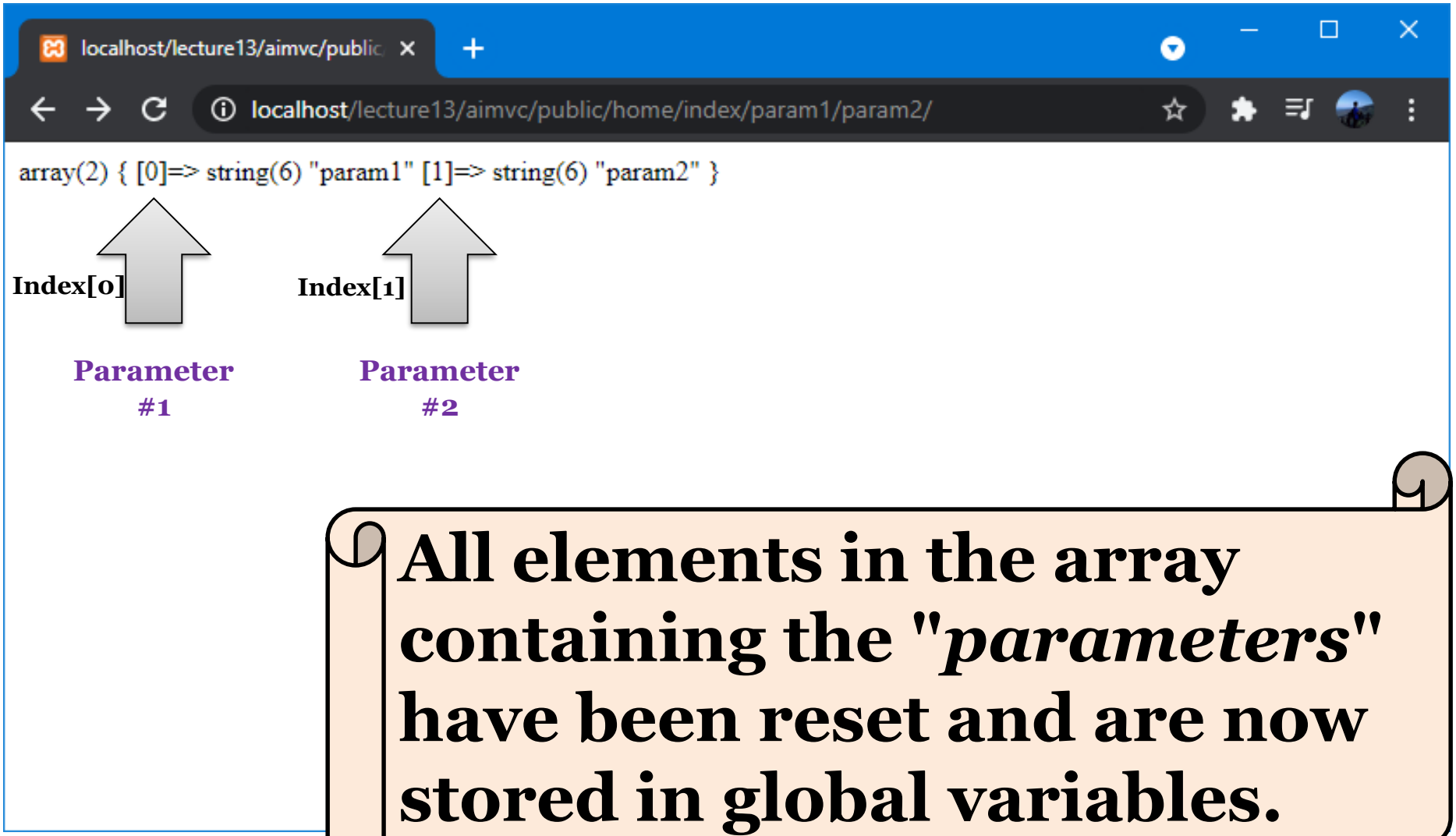
Element of *index 1* (value: *index*) already recognized as a *default method name* (*index*).

6 – Handle URL Request with ***Parameter***

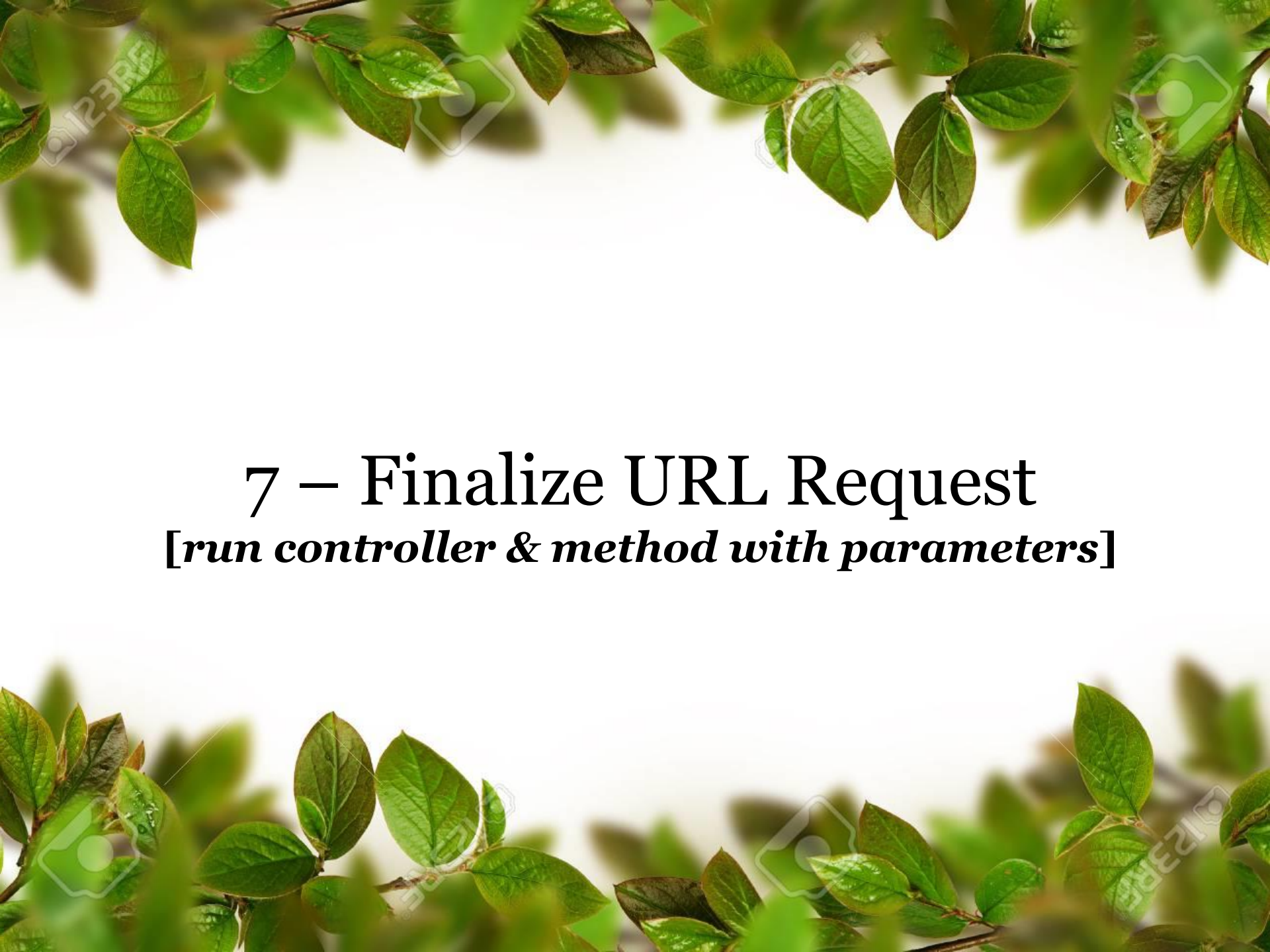
Recognize Input Parameters



Test Handle Input Parameters



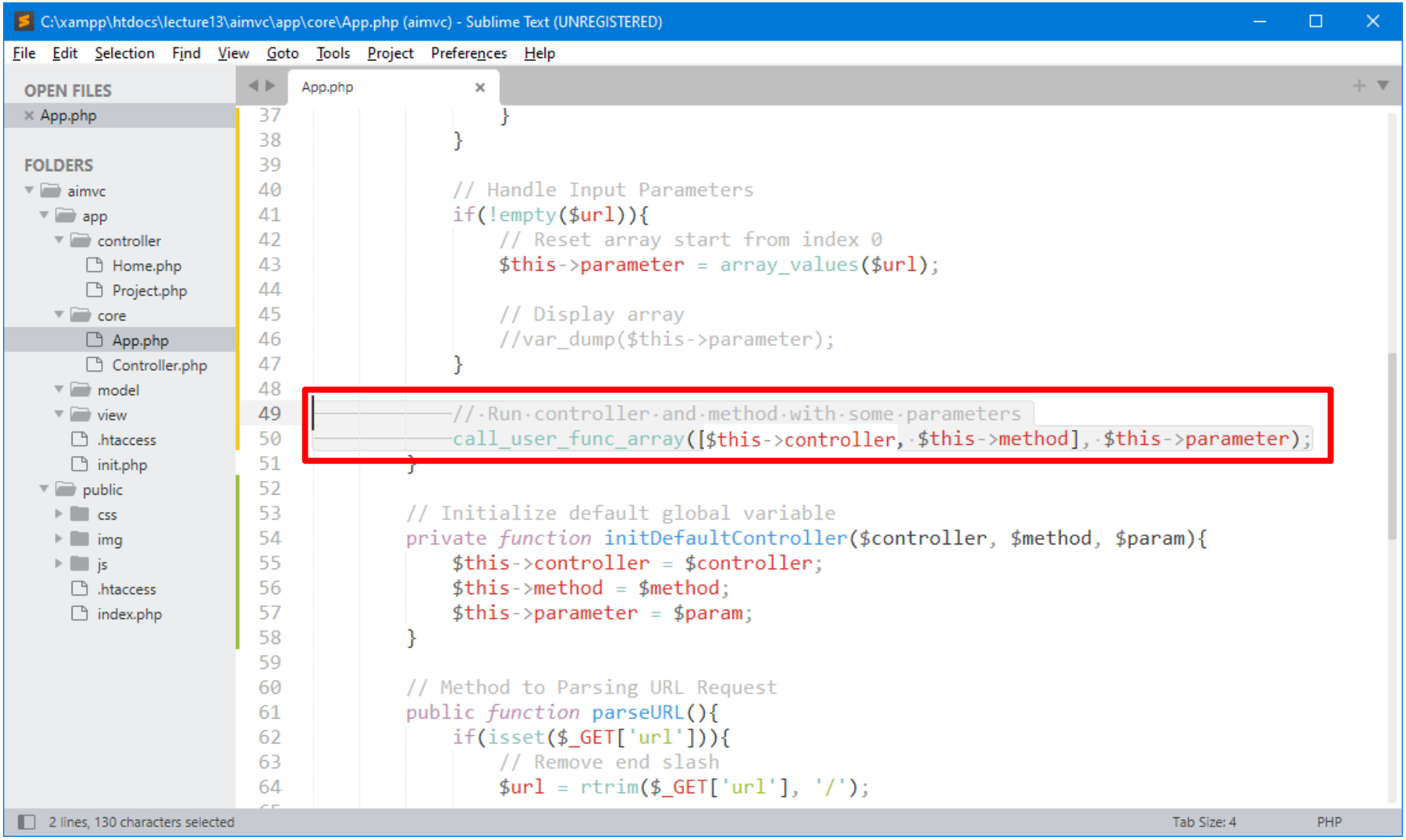
All elements in the array containing the "*parameters*" have been reset and are now stored in global variables.

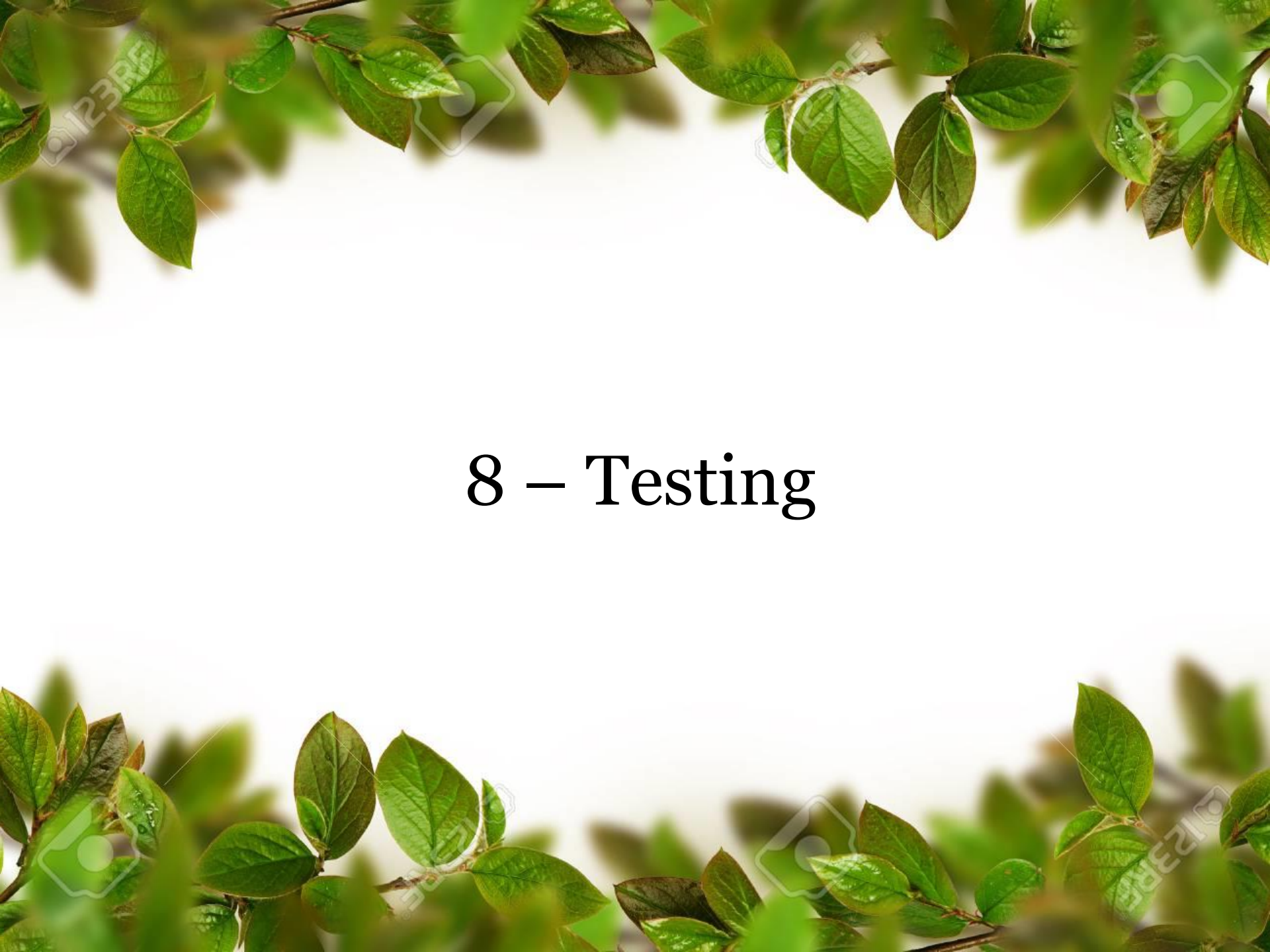


7 – Finalize URL Request

[*run controller & method with parameters*]

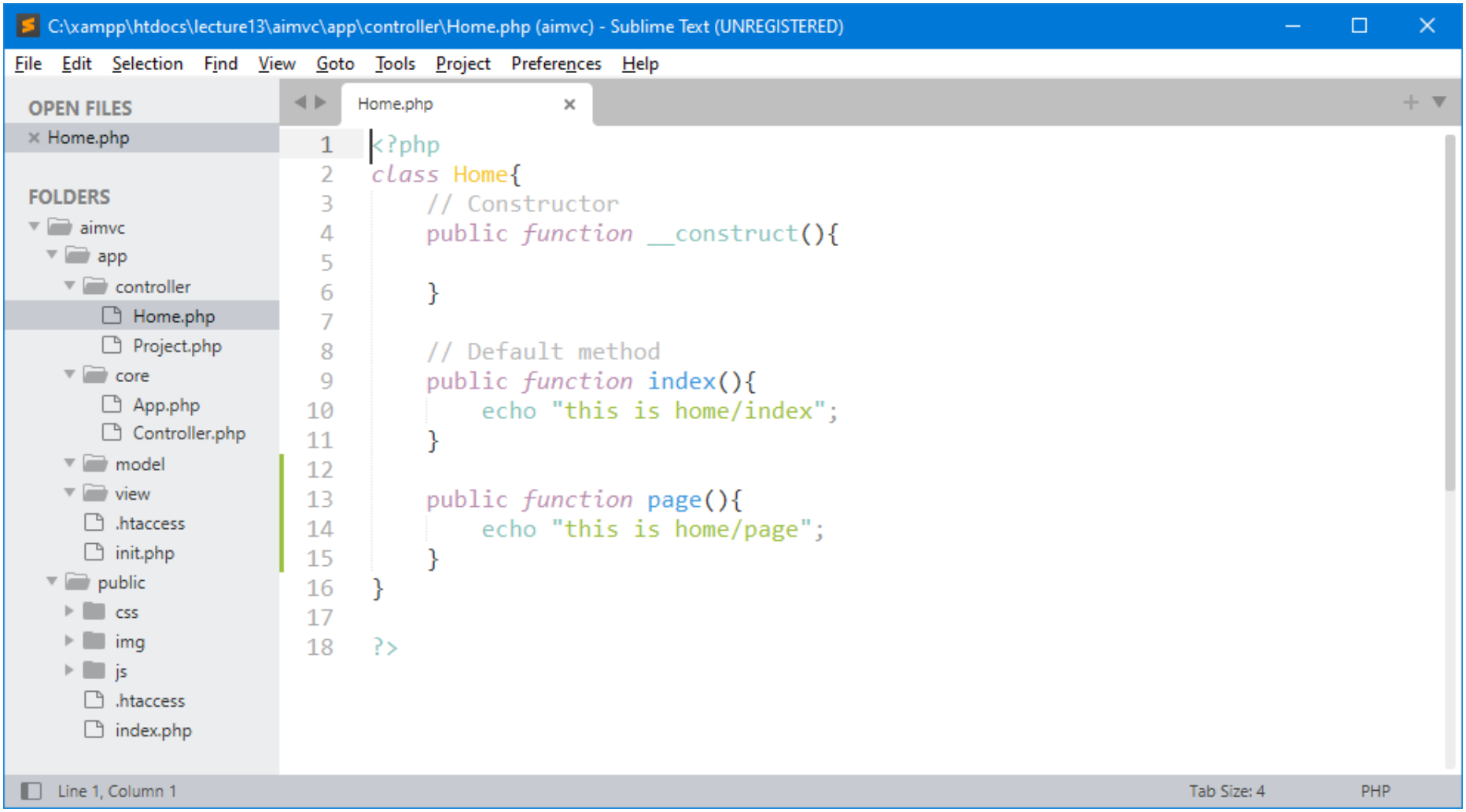
Run controller & method



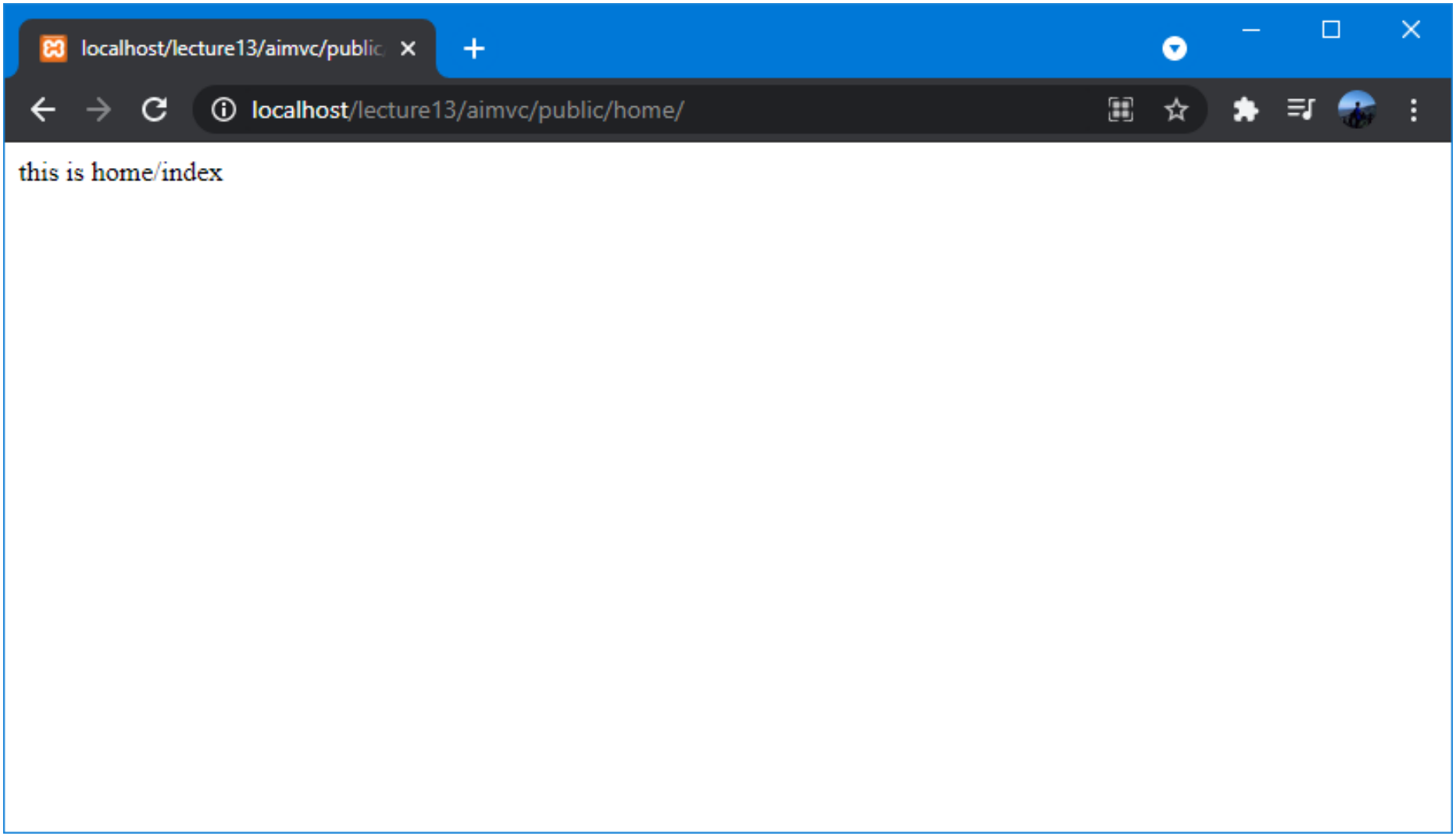


8 – Testing

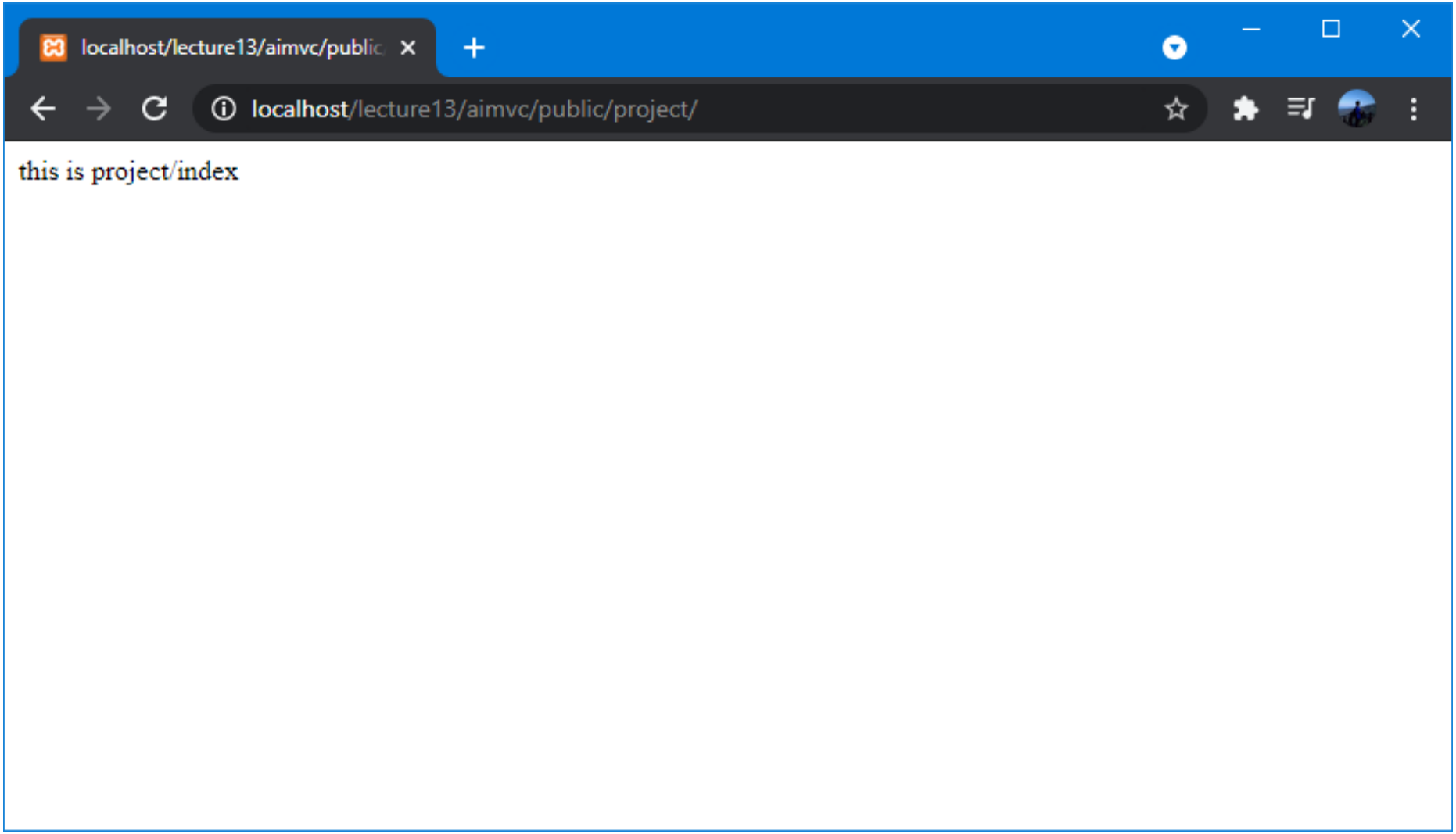
Add More Methods



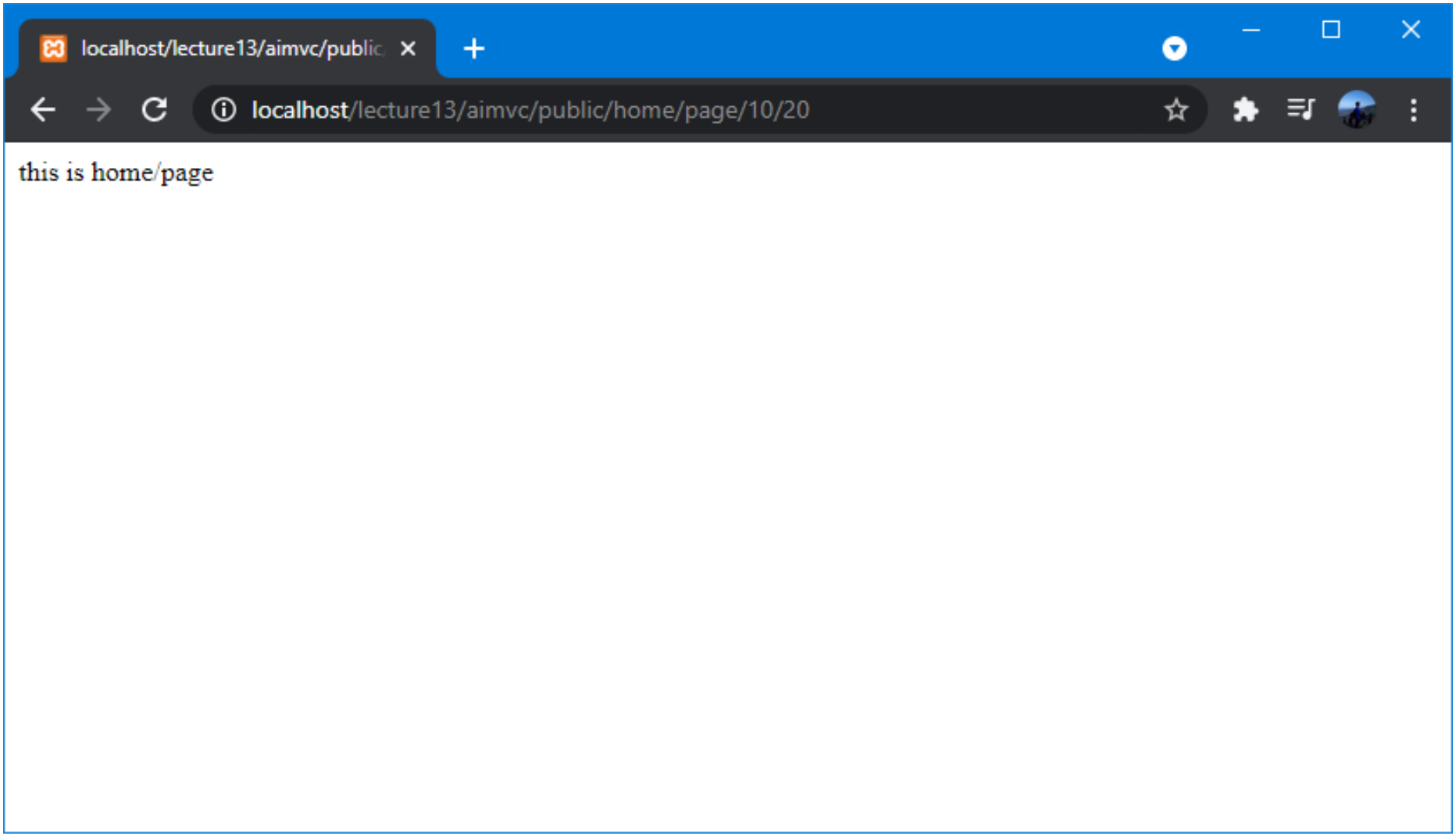
Access Home Controller (*index*)



Access Project Controller (*index*)



Access Home Controller with Specific Method



exercise for students

Latihan untuk *Students*

Create directory : C:\xampp\htdocs\lecture13\
Create file : _.php

Buat semua contoh yang sudah di pelajari sebelumnya.

Reference (#1)

Controllers
https://codeigniter.com/userguide3/general/controllers.html#id1

URI Routing
https://codeigniter.com/userguide3/general/routing.html

Model View Controller example
https://developer.mozilla.org/en-US/docs/Glossary/MVC

Learn About MVC Architecture
https://www.c-sharpcorner.com/article/learn-about-mvc-architecture/

MVC URL Routing
https://sudhersonv.wixsite.com/codeblogs/single-post/2015/05/19/mvc-url-routing

Model View Controller Pattern – MVC Architecture and Frameworks
https://www.freecodecamp.org/news/the-model-view-controller-pattern-mvc-architecture-and-frameworks-explained/

Apa itu MVC? Mengenal Mvc (Model View Controller)
https://note-ade.blogspot.com/2020/10/mvc.html

Sistem Pemrograman Model View Controller (MVC)
https://informatika.uc.ac.id/id/2016/12/sistem-pemrograman-model-view-controller-mvc/

A Beginner's guide to Ruby on Rails MVC
https://hackernoon.com/beginners-guide-to-ruby-on-rails-mvc-model-view-controller-pattern-4z19196a

END PRESENTATION

Thank you for your attention

Instructor: S – W – T

THANK YOU